Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/27/2014
PI Name:	Muday, Gloria Ph.D.		
Project Title:	Dynamics of auxin transport protein localization and gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	08/01/2009	**End Date:**	01/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 1/31/2014, per NSSC information (Ed., 7/25/13) NOTE: Extended to 7/31/2013, per NSSC information (Ed., 12/18/2012) NOTE: extended to 12/31/2012 per NSSC information (Ed., 4/23/2012)		
Key Personnel Changes/Previous PI:			
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Performance Goal Text:			

Task Description:	Plants respond to gravity in numerous ways; the best understood is the change in growth orientation in response to reorientation of plants relative to the gravity vector. A growth asymmetry develops, which allows the plant organ to return to the original angle of growth relative to gravity. Understanding gravitropism is essential to facilitate growth of plants in the microgravity environment of space. Gravitropic bending is mediated, at least in part, by redistribution of the plant hormone auxin. This proposal examines the mechanisms by which the cellular asymmetry of auxin transport proteins is established and the molecular mechanisms that allow translocation of the auxin signals from the site of perception to the site of reaction. We will employ three approaches to transiently alter the expression and localization of auxin transport proteins to examine the effect on gravitropic curvature in roots of the model species Arabidopsis thaliana. First, we will use a temperature sensitive mutant, scd1-1, with a defect in endomembrane targeting to transiently disrupt auxin transport protein localization and gravitropic curvature. Second, we will transiently perturb the activity and/or localization of auxin transport proteins through a pharmacological approach and measure the resulting effect on gravitropic kinetics. In our third approach, we will use well characterized null mutants with defects in auxin transport proteins and will express wild type copies of defective genes in an inducible expression system. This approach of reintroducing a gene function at a controlled time point during the gravitropic process, should allow us to understand when these proteins function in controlling auxin transport and gravitropism. The planned experiments will combine high resolution laser scanning confocal microscopy and high temporal resolution gravitropic analyses with transient perturbation of auxin transport to understand the dynamics of auxin transport protein localization and resulting auxin asymmetries that drive differential gravitropic growth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have now completed an analysis of the gravitropic and auxin transport phenotypes of the temperature sensitive scd1-1 mutant, which has a defect in a gene encoding a protein predicted to function in Rab GTPase-dependent protein trafficking. All aspects of root growth and gravitropism are normal in this mutant at the permissive temperature of 18°C, while at the nonpermissive temperature of 25°C, scd1-1 exhibits significant reductions in gravitropic curvature, growth rate, and auxin transport. Basipetal or shootward IAA transport is reduced in the scd1-1 mutant relative to wild-type at 25°C, while similar transport amounts are detected at 18°C. These differences were observed using both traditional radioactive IAA transport assay and by applying auxin at the root tip and measurement of the shootward progression of fluorescent signal resulting from auxin induced gene expression, using the DR5:GFP reporter. Just 8 hours of incubation at RT (25°C) after growth at 18°C is sufficient to show a significant reduction in IAA transport, nicely mirroring the timing of the defect in gravitropism. The final line of experiments in scd1-1 focuses on examination of the effects of temperature changes on the distribution of auxin transport proteins in this mutant background. We have used laser scanning confocal microscopy to examine the tissue level and sub-cellular distribution of auxin efflux carriers PIN1, PIN2, PIN3, AUX1, and ABCB19 that are translational fusions to GFP, with PIN2:GFP being best linked to gravity response. The pattern and intensity of fluorescence was similar in wild-type and scd1-1 at 18 degrees C. At 25°C, the PIN1 and PIN2:GFP fusion in scd1-1 were found at substantially lower levels in the plasma membrane, and accumulated in endomembranes, while changes in the other reporters were subtle. The membrane asymmetries of PIN1:GFP and PIN2:GFP reporters were generally maintained, although the membrane associated fluorescence signal was substantially weaker. The apparent structure and character of endomembranes that accumulated PIN1-GFP and PIN2-GFP were quite different. This result suggests that in the mutant, these two auxin transport proteins may be trafficked by different pathways in the different cell types in which they are found, causing them to be shunted to different structures where they accumulate. The final step of these experiments has been the identification of the nature of the endomembranes that accumulate PIN2:GFP. We have crossed endomembrane specific fluorescent markers for early and late endosomes, Golgi, endoplasmic reticulum, and vacuole into scd1-1xPIN1 GFP and scd1-1xPIN2GFP. Our studies indicate that PIN2:GFP accumulates in the late endosomes in scd1-1, consistent with a defect in a RAB-GEF that acts at this final stage of targeting. We are currently completing the manuscript describing these results. A second line of experimentation utilized a pharmacological approach to transiently disrupt activity and/or targeting of auxin transport proteins to understand their role in gravitropic responses. We have examined compounds which inhibit protein phosphorylation and dephosphorylation, specifically staurosporine and cantharidin, respectively, and combined these inhibitors with mutants in specific kinase and phosphatase genes, to provide evidence of reciprocal regulation of auxin transport by protein phosphorylation and regulation of gravitropic responses (Sukumar et al., 2009). An undergraduate student completed a study of three endosidin compounds, ES10, ES11, and ES12, on gravity response and auxin transport. These endosidin chemicals were isolated in a chemical genomic screen by Glenn Hicks and Natasha Raikhel at UC Riverside for inhibition of hypocotyl gravitropism. These experiments quantified the kinetics of inhibition of root gravitropism by these three compounds using high resolution imaging and identified doses of compounds that inhibited gravitropism, but not root elongation. ES11 and ES12, but not ES10, perturb the formation of asymmetric gradients of auxin induced gene expression across a gravity stimulated root. The effect of these compounds on auxin transport was quantified and only ES12 reduced shootward or basipetal IAA polar transport, which supplies auxin for root gravitropic curvature. Therefore these three compounds fall into three distinct activity pathways, suggesting they act on different protein targets. The effect of a fourth endosidin, ES2, on gravitropism and auxin transport was examined initially by Jonathan Watkins, and is now being examined in ES2 resistant mutants isolated from Natasha Raikhel's lab, as the final step in publication of an ES2 focused manuscript.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Articles in Peer-reviewed Journals	Watkins JM, Hechler PJ, Muday GK. "Ethylene-induced flavonol accumulation in guard cells suppresses reactive oxygen species and modulates stomatal aperture." Plant Physiology. 2014 Apr;164(4):1707-17. http://dx.doi.org/10.1104/pp.113.233528 ; PubMed PMID: 24596331; PubMed Central PMCID: PMC3982735 , Apr-2014

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Articles in Peer-reviewed Journals	Gerttula S, Zinkgraf M, Muday G, Lewis D, Ibatullin FM, Brumer H, Hart F, Mansfield SD, Filkov V, Groover A. "Transcriptional and hormonal regulation of gravitropism of woody stems in Populus." Plant Cell. 2015 Oct;27(10):2800-13. Epub 2015 Sep 26. http://dx.doi.org/10.1105/tpc.15.00531 ; PubMed PMID: 26410302 , Oct-2015
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Awards	Isley J. "Second Place Graduate student poster award. 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010." Nov-2010
Awards	Hughes R. "1st Place Undergraduate Poster Presentations, 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010." Nov-2010
Dissertations and Theses	Mattox C. "Examination of auxin transport and root development in the scd1-1 mutant in Arabidopsis thaliana." MS thesis, Wake Forest University, May 2009. , May-2009
Journal/Magazine covers	Gerttula S, Zinkgraf M, Muday G, Lewis D, Ibatullin FM, Brumer H, Hart F, Mansfield SD, Filkov V, Groover A. "Cover in journal Plant Cell for article, 'Transcriptional and hormonal regulation of gravitropism of woody stems in Populus.' " Plant Cell. 2015 Oct;27(10):2800-13. Epub 2015 Sep 26. http://dx.doi.org/10.1105/tpc.15.00531 ; PubMed PMID: 26410302 ; http://www.plantcell.org/content/27/10.cover-expansion , Oct-2015